DAILY PRACTICE CALENDAR

AP Calculus BC • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **AP Calculus BC** course framework (AP Units 1–10). AB topics are reviewed at a faster pace to leave time for the BC-only topics: integration techniques, improper integrals, logistic models, parametric, polar and vector functions, and infinite series. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Limits & Continuity Review
2	Aug 23–Aug 27	Derivative Rules
3	Aug 30–Sep 3	Chain Rule, Implicit & Inverse Derivatives
4	Sep 6–Sep 10	Related Rates & Linearization
5	Sep 13–Sep 17	L'Hôpital's Rule & the Mean Value Theorem
6	Sep 20–Sep 24	Extrema & Concavity
7	Sep 27–Oct 1	Optimization & Motion
8	Oct 4–Oct 8	Riemann Sums & the Fundamental Theorem
9	Oct 11–Oct 15	Antiderivatives & Substitution
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Integration by Parts
12	Nov 1–Nov 5	Partial Fractions
13	Nov 8–Nov 12	Improper Integrals
14	Nov 15–Nov 19	Slope Fields & Euler's Method
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Separable & Logistic Differential Equations
16	Dec 6–Dec 10	Average Value & Accumulation
17	Dec 13–Dec 17	Area & Volume
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Washers & Arc Length
19	Jan 10–Jan 14	Motion & Accumulation in Context
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Parametric Derivatives
22	Jan 31–Feb 4	Parametric Arc Length & Motion
23	Feb 7–Feb 11	Vector-Valued Functions
24	Feb 14–Feb 18	Polar Curves & Derivatives
25	Feb 21–Feb 25	Area in Polar Coordinates
26	Feb 28–Mar 3	Sequences & Geometric Series
27	Mar 6–Mar 10	nth-Term, Integral & p-Series Tests
28	Mar 13–Mar 17	Comparison & Ratio Tests
29	Mar 20–Mar 24	Alternating Series & Error Bounds
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Taylor & Maclaurin Polynomials
31	Apr 10–Apr 14	Power Series
32	Apr 17–Apr 21	AP Review: Derivatives & Integrals
33	Apr 24–Apr 28	AP Review & Practice Exam
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Limits & Derivatives

Limits & Continuity Review

Core: Substitute first; if you get $\frac{0}{0}$, factor, rationalize or use a known limit.

Mon Aug 16
Algebraic limits
TRY Find $\lim_{x\to 4}\frac{x^2-16}{x-4}$.

Tue Aug 17
Trig limits
TRY Find $\lim_{x\to 0}\frac{\sin 5x}{2x}$.

Wed Aug 18
Continuity
TRY $f(x)=ax^2$ for $x<1$ and $3x+1$ for $x\geq 1$. Find a so f is continuous.

Thu Aug 19
Limits at infinity
TRY Find $\lim_{x\to\infty}\frac{2x+1}{\sqrt{x^2+3}}$.

Fri Aug 20
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Limits & Derivatives

Derivative Rules

Core: Power, product and quotient rules, plus the basic trig and exponential derivatives.

Mon Aug 23
Power rule
TRY Find $\frac{d}{dx}\left(x^3-\frac{4}{x}\right)$.

Tue Aug 24
Product rule
TRY Find $\frac{d}{dx}\left(x^2\sin x\right)$.

Wed Aug 25
Quotient rule
TRY Find $\frac{d}{dx}\frac{e^x}{x}$.

Thu Aug 26
Tangent lines
TRY Find the tangent line to $y=x^3$ at $x=2$.

Fri Aug 27
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Limits & Derivatives

Chain Rule, Implicit & Inverse Derivatives

Core: Outside, then inside; every y gets a $\frac{dy}{dx}$.

Mon Aug 30
Chain rule
TRY Find $\frac{d}{dx}\cos(3x^2)$.

Tue Aug 31
Log of a trig function
TRY Find $\frac{d}{dx}\ln(\sec x)$.

Wed Sep 1
Implicit differentiation
TRY Find $\frac{dy}{dx}$ for $x^2+x\cdot y+y^2=7$ at $(1,2)$.

Thu Sep 2
Inverse trig
TRY Find $\frac{d}{dx}\arcsin(2x)$.

Fri Sep 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Limits & Derivatives

Related Rates & Linearization

Core: Link the quantities, then differentiate with respect to time.

Mon Sep 6
HOLIDAY: Labor Day
No class. Do this skill's 10 problems on the weekend:
Spheres
TRY A balloon's radius grows at 0.5 cm/s. How fast is its volume growing when $r=10$?

Tue Sep 7
Ladders
TRY A 13-ft ladder's foot slides out at 2 ft/s. How fast is the top falling when the foot is 5 ft from the wall?

Wed Sep 8
Linear approximation
TRY Use a tangent line to estimate $\sqrt[3]{8.3}$.

Thu Sep 9
Over or under?
TRY Is that estimate too high or too low?

Fri Sep 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Applying Derivatives & Integrals

L'Hôpital's Rule & the Mean Value Theorem — **Core:** Use L'Hôpital only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ forms.

Mon Sep 13
L'Hôpital twice
TRY Find $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

Tue Sep 14
Indeterminate powers
TRY Find $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x$.

Wed Sep 15
Mean Value Theorem
TRY For $f(x) = x^3$ on $[0, 3]$, find the c given by the MVT.

Thu Sep 16
Rolle's theorem
TRY For $f(x) = x^2 - 4x$ on $[0, 4]$, find c with $f'(c) = 0$.

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Applying Derivatives & Integrals

Extrema & Concavity — **Core:** Use f' for increasing and decreasing, f'' for concavity.

Mon Sep 20
Critical points
TRY Find the critical points of $f(x) = x^4 - 8x^2$.

Tue Sep 21
Classifying
TRY For the same f, is $x = 0$ a local max or a local min?

Wed Sep 22
Inflection points
TRY Find the inflection points of $f(x) = x^4 - 8x^2$.

Thu Sep 23
Absolute extrema
TRY Find the absolute max and min of $f(x) = x \cdot e^{-x}$ on $[0, 3]$.

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Applying Derivatives & Integrals

Optimization & Motion — **Core:** Velocity is s'; speed is $|v|$; total distance adds up every change of direction.

Mon Sep 27
Optimization
TRY A closed box with a square base has volume 1,000. What is its least possible surface area?

Tue Sep 28
Turning around
TRY For $s(t) = t^3 - 9t^2 + 24t$, when is the velocity 0?

Wed Sep 29
Speed
TRY Find the speed at $t = 3$.

Thu Sep 30
Total distance
TRY Find the total distance traveled on $[0, 4]$.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Applying Derivatives & Integrals

Riemann Sums & the Fundamental Theorem — **Core:** A definite integral is a limit of Riemann sums and equals $F(b) - F(a)$.

Mon Oct 4
Midpoint sums
TRY Estimate $\int_0^4 x^2\,dx$ with 2 equal midpoint rectangles.

Tue Oct 5
FTC
TRY Find $\int_0^{\pi/2} \cos x\,dx$.

Wed Oct 6
FTC with the chain rule
TRY Find $\frac{d}{dx}\int_1^{3x} e^{t^2}\,dt$.

Thu Oct 7
Limits of sums
TRY Find $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{n}\left(\frac{k}{n}\right)^2$.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Applying Derivatives & Integrals

Antiderivatives & Substitution — **Core:** Choose u so that du is already in the integrand.

Mon Oct 11
Basic antiderivatives
TRY Find $\int \left(3x^2 - \frac{2}{x}\right) dx$.

Tue Oct 12
Substitution
TRY Find $\int \sin x \cos x \, dx$.

Wed Oct 13
Log results
TRY Find $\int \frac{x}{x^2+1} \, dx$.

Thu Oct 14
Definite substitution
TRY Find $\int_0^{\ln 2} \frac{e^x}{1+e^x} \, dx$.

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Applying Derivatives & Integrals

Review & Checkpoint 1 — **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: limits
TRY Find $\lim_{x \to 0} \frac{e^{2x}-1}{x}$.

Tue Oct 19
Review: derivatives
TRY Find $\frac{d}{dx}(x \ln x)$.

Wed Oct 20
Review: concavity
TRY Is $y = \ln x$ concave up or concave down?

Thu Oct 21
Review: integrals
TRY Find $\int_1^2 \frac{1}{x^2} \, dx$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Integration Techniques & Differential Equations

Integration by Parts — **Core:** $\int u \, dv = uv - \int v \, du$: let u be the part that gets simpler.

Mon Oct 25
Polynomial × exponential
TRY Find $\int x \cdot e^x \, dx$.

Tue Oct 26
Polynomial × trig
TRY Find $\int x \cos x \, dx$.

Wed Oct 27
Logarithms
TRY Find $\int \ln x \, dx$.

Thu Oct 28
Definite integrals
TRY Find $\int_0^1 x \cdot e^x \, dx$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Integration Techniques & Differential Equations

Partial Fractions — **Core:** Split a rational function into simpler fractions with linear denominators.

Mon Nov 1
Decomposing
TRY Write $\frac{1}{x(x+1)}$ as a sum of simpler fractions.

Tue Nov 2
Integrating
TRY Find $\int \frac{1}{x(x+1)} \, dx$.

Wed Nov 3
Another one
TRY Find $\int \frac{5}{(x-1)(x+4)} \, dx$.

Thu Nov 4
Definite integrals
TRY Find $\int_1^2 \frac{1}{x(x+1)} \, dx$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Integration Techniques & Differential Equations

Improper Integrals

Core: Replace the bad bound with a variable and take a limit.

Mon Nov 8
Infinite bounds
TRY Find $\int_1^\infty \frac{1}{x^2}\,dx$.

Tue Nov 9
Divergence
TRY Does $\int_1^\infty \frac{1}{x}\,dx$ converge?

Wed Nov 10
Exponential decay
TRY Find $\int_0^\infty e^{-x}\,dx$.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Unbounded integrands
TRY Find $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Integration Techniques & Differential Equations

Slope Fields & Euler's Method

Core: Euler's method: new y = old y + (slope) $\times$ (step).

Mon Nov 15
Slope fields
TRY For $\frac{dy}{dx} = x + y$, find the slope at $(1, -1)$.

Tue Nov 16
One Euler step
TRY For $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use a step of 0.5 to estimate $y(0.5)$.

Wed Nov 17
Two Euler steps
TRY Take one more step to estimate $y(1)$.

Thu Nov 18
Over or under?
TRY If the solution curve is concave up, is Euler's estimate too high or too low?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Integration Techniques & Differential Equations

Separable & Logistic Differential Equations

Core: Logistic growth: $\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$, with carrying capacity L.

Mon Nov 29
Separable equations
TRY Solve $\frac{dy}{dx} = y^2$ with $y(0) = 1$.

Tue Nov 30
Exponential decay
TRY Solve $\frac{dy}{dt} = -0.2y$ with $y(0) = 50$.

Wed Dec 1
Carrying capacity
TRY For $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$, find the carrying capacity.

Thu Dec 2
Fastest growth
TRY For the same model, at what population is P growing fastest?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Applications of Integration

Average Value & Accumulation

Core: Final amount = starting amount + the integral of the rate.

Mon Dec 6
Average value
TRY Find the average value of $\sin x$ on $[0, \pi]$.

Tue Dec 7
Accumulation
TRY Cars pass a point at $r(t) = 3t^2$ cars per hour. How many pass from $t = 1$ to $t = 3$?

Wed Dec 8
Displacement & distance
TRY For $v(t) = \cos t$ on $[0, \pi]$, find the displacement and the total distance.

Thu Dec 9
From a starting value
TRY If $f(2) = 5$ and $f'(x) = 2x$, find $f(5)$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Applications of Integration

Area & Volume **Core:** Area: top minus bottom. Volume: add up the cross-section areas.

Mon Dec 13
Area between curves
TRY Find the area between $y = x^2$ and $y = 2 - x^2$.

Tue Dec 14
Integrating in y
TRY Find the area between $x = y^2$ and $x = y + 2$.

Wed Dec 15
Discs
TRY Rotate the region under $y = e^x$ on $[0, 1]$ about the x-axis. Find the volume.

Thu Dec 16
Cross sections
TRY The base is between $y = 1 - x^2$ and the x-axis. Cross sections perpendicular to the x-axis are squares. Find the volume.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Applications of Integration

Washers & Arc Length **Core:** Washer: $\pi \int (R^2 - r^2)\, dx$. Arc length: $\int \sqrt{1 + (y')^2}\, dx$.

Mon Jan 3
Washers
TRY Rotate the region between $y = \sqrt{x}$ and $y = x$ about the x-axis. Find the volume.

Tue Jan 4
Another axis
TRY Rotate the region under $y = x^2$ on $[0, 1]$ about the line $y = -1$. Find the volume.

Wed Jan 5
Arc length
TRY Find the length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

Thu Jan 6
Setting up
TRY Write the integral for the length of $y = x^2$ on $[0, 1]$.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Applications of Integration

Motion & Accumulation in Context **Core:** Explain every answer with units and the meaning of the integral.

Mon Jan 10
Position
TRY If $v(t) = 2t - 4$ and $s(0) = 3$, find $s(5)$.

Tue Jan 11
Total distance
TRY For the same $v(t)$, find the total distance on $[0, 5]$.

Wed Jan 12
Largest amount
TRY Water enters at $r(t) = 20 - t$ L/min and leaves at 10 L/min. When is the amount greatest?

Thu Jan 13
Meaning
TRY $R(t)$ is gallons per minute. What does $\int_0^{60} R(t)\, dt$ mean?

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Applications of Integration

Mid-Year Review & Checkpoint **Core:** Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: integration by parts
TRY Find $\int x \sin x\, dx$.

Tue Jan 18
Review: partial fractions
TRY Find $\int \frac{2}{x^2 - 1}\, dx$.

Wed Jan 19
Review: improper integrals
TRY Find $\int_1^{\infty} \frac{1}{x^3}\, dx$.

Thu Jan 20
Review: logistic models
TRY Find the carrying capacity of $\frac{dP}{dt} = 2P\left(1 - \frac{P}{80}\right)$.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028

Unit 5: Parametric, Polar & Vector Functions

Parametric Derivatives

Core: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

Mon Jan 24
First derivative
TRY For $x = t^2$ and $y = t^3$, find $\frac{dy}{dx}$.

Tue Jan 25
Tangent lines
TRY Find the tangent line to that curve at $t = 2$.

Wed Jan 26
Second derivative
TRY Find $\frac{d^2y}{dx^2}$ for the same curve.

Thu Jan 27
Horizontal tangents
TRY For $x = t^2 - 1$ and $y = t^3 - 3t$, where are the tangents horizontal?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028

Unit 5: Parametric, Polar & Vector Functions

Parametric Arc Length & Motion

Core: Speed $= \sqrt{(x')^2 + (y')^2}$; arc length is the integral of speed.

Mon Jan 31
Speed
TRY Find the speed for $x = 3t$, $y = 4t$.

Tue Feb 1
Arc length of a circle
TRY Find the length of $x = \cos t$, $y = \sin t$ for $0 \le t \le \pi$.

Wed Feb 2
Arc length
TRY Find the length of $x = t^2$, $y = \frac{2}{3}t^3$ for $0 \le t \le \sqrt{3}$.

Thu Feb 3
Distance vs. displacement
TRY For $x = \cos t$, $y = \sin t$ on $[0, 2\pi]$, find the distance traveled and the displacement.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028

Unit 5: Parametric, Polar & Vector Functions

Vector-Valued Functions

Core: Differentiate and integrate each component separately.

Mon Feb 7
Velocity
TRY For $\mathbf{r}(t) = \langle t^2, e^t \rangle$, find $\mathbf{v}(t)$.

Tue Feb 8
Speed
TRY Find the speed for $\mathbf{r}(t) = \langle 3\cos t, 3\sin t \rangle$.

Wed Feb 9
Acceleration
TRY For $\mathbf{r}(t) = \langle t^3, 2t \rangle$, find $\mathbf{a}(1)$.

Thu Feb 10
Position from velocity
TRY If $\mathbf{v}(t) = \langle 2t, 1 \rangle$ and $\mathbf{r}(0) = \langle 1, 3 \rangle$, find $\mathbf{r}(2)$.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028

Unit 5: Parametric, Polar & Vector Functions

Polar Curves & Derivatives

Core: Use $x = r\cos\theta$ and $y = r\sin\theta$, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$.

Mon Feb 14
Polar to rectangular
TRY Write $r = 2\sin\theta$ in rectangular form.

Tue Feb 15
Meaning of $\frac{dr}{d\theta}$
TRY For $r = 3 + 2\cos\theta$, is the curve moving toward or away from the origin at $\theta = \frac{\pi}{2}$?

Wed Feb 16
Polar slopes
TRY For $r = \theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$.

Thu Feb 17
Through the pole
TRY For which θ in $[0, \pi)$ does $r = \cos 2\theta$ pass through the origin?

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Parametric, Polar & Vector Functions

Area in Polar Coordinates

Core: Area $= \frac{1}{2}\int_{\alpha}^{\beta} r^2\,d\theta$.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Circles

TRY Use the polar area formula to find the area inside $r = 3$.

Tue Feb 22

Cardioids

TRY Find the area inside $r = 1 + \cos\theta$.

Wed Feb 23

One petal

TRY Find the area of one petal of $r = \sin 2\theta$.

Thu Feb 24

Between curves

TRY Find the area inside $r = 2$ and outside $r = 1$.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 6: Infinite Series

Sequences & Geometric Series

Core: A geometric series with $|r| < 1$ adds to $\frac{\text{first term}}{1-r}$.

Mon Feb 28

Sequences

TRY Find $\lim_{n\to\infty} \frac{3n+1}{n-2}$.

Tue Feb 29

Geometric series

TRY Find $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$.

Wed Mar 1

Watch the start

TRY Find $\sum_{n=1}^{\infty} 5\left(\frac{1}{4}\right)^n$.

Thu Mar 2

Telescoping

TRY Find $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Infinite Series

nth-Term, Integral & p-Series Tests

Core: If the terms do not go to 0, the series diverges; $\sum \frac{1}{n^p}$ converges only when $p > 1$.

Mon Mar 6

nth-term test

TRY Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge?

Tue Mar 7

p-series

TRY Does $\sum_{n=1}^{\infty} \frac{1}{n^{1.5}}$ converge?

Wed Mar 8

The harmonic series

TRY Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge?

Thu Mar 9

Integral test

TRY Does $\sum_{n=2}^{\infty} \frac{1}{n\ln n}$ converge?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Infinite Series

Comparison & Ratio Tests

Core: Compare with a known series, or check $\lim \left|\frac{a_{n+1}}{a_n}\right|$.

Mon Mar 13

Direct comparison

TRY Does $\sum_{n=1}^{\infty} \frac{1}{n^2+5}$ converge?

Tue Mar 14

Limit comparison

TRY Does $\sum_{n=1}^{\infty} \frac{1}{2n+3}$ converge?

Wed Mar 15

Ratio test

TRY Does $\sum_{n=1}^{\infty} \frac{n!}{10^n}$ converge?

Thu Mar 16

Ratio test again

TRY Does $\sum_{n=0}^{\infty} \frac{3^n}{n!}$ converge?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Infinite Series

Alternating Series & Error Bounds — **Core:** For an alternating series, the error is at most the first unused term.

Mon Mar 20
Alternating series test
TRY Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge? Absolutely or conditionally?

Tue Mar 21
Absolute convergence
TRY Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ converge absolutely?

Wed Mar 22
Error bound
TRY Using 4 terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, what is the error bound?

Thu Mar 23
Alternating geometric series
TRY Find $\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n$.

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Infinite Series

Taylor & Maclaurin Polynomials — **Core:** The coefficient of $(x-a)^n$ is $\frac{f^{(n)}(a)}{n!}$.

Mon Apr 3
Maclaurin polynomials
TRY Write the first four terms of the Maclaurin series for e^x.

Tue Apr 4
Taylor polynomials
TRY Find the 3rd-degree Taylor polynomial of $\ln x$ about $x = 1$.

Wed Apr 5
Approximating
TRY Use $1 - \frac{x^2}{2}$ to estimate $\cos 0.1$.

Thu Apr 6
Lagrange error
TRY Bound the error when $P_2(x) = 1 + x + \frac{x^2}{2}$ estimates $e^{0.1}$. (Use $e^{0.1} < 1.2$.)

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Infinite Series

Power Series — **Core:** Use the ratio test for the radius, then test each endpoint separately.

Mon Apr 10
Radius of convergence
TRY Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{3^n}$.

Tue Apr 11
Interval of convergence
TRY Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

Wed Apr 12
Known series
TRY Write the Maclaurin series for $\sin x$.

Thu Apr 13
New series from old
TRY Write the first three nonzero terms of the series for $x \cdot e^{x^2}$.

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — AP Exam Review

AP Review: Derivatives & Integrals — **Core:** Mix the skills from Units 1–4 and justify each answer.

Mon Apr 17
Review: L'Hôpital
TRY Find $\lim_{x \to 0} \frac{\tan x - x}{x^3}$.

Tue Apr 18
Review: logarithmic differentiation
TRY Find $\frac{d}{dx} x^x$.

Wed Apr 19
Review: parts twice
TRY Find $\int x^2 e^x \, dx$.

Thu Apr 20
Timed set: 10 in 20 min
TRY Find $\int_0^{\infty} x \cdot e^{-x} \, dx$.

Fri Apr 21
Weekly Quiz
10 mixed AP-style problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — AP Exam Review

AP Review & Practice Exam

Core: Practice a full exam, then learn from every error.

Mon Apr 24

Review: series

TRY Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converge absolutely, conditionally, or not at all?

Tue Apr 25

Review: parametric

TRY For $x = 2t$ and $y = t^2$, find $\frac{dy}{dx}$ at $t = 3$.

Wed Apr 26

Review: polar area

TRY Find the area inside $r = 2\sin\theta$.

Thu Apr 27

Practice Exam

Full-length AP Calculus BC practice exam, timed.

Fri Apr 28

Error-log review

Redo every missed problem from the exam; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — AP Exam Review

Final Checkpoint & Celebration

Core: Show what you have learned and go into the AP exam with confidence.

Mon May 1

Light review

TRY Find $\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$.

Tue May 2

Mixed warm-up set

TRY Find $\int_0^1 2x\,dx$.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for college math: Multivariable Calculus and Linear Algebra.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 8
Tue: $\frac{5}{2}$
Wed: $a = 4$
Thu: 2

Week 2
Mon: $3x^2 + \frac{4}{x^2}$
Tue: $2x \sin x + x^2 \cos x$
Wed: $\frac{e^x(x-1)}{x^2}$
Thu: $y = 12x - 16$

Week 3
Mon: $-6x \sin(3x^2)$
Tue: $\tan x$
Wed: $-\frac{4}{5}$
Thu: $\frac{2}{\sqrt{1-4x^2}}$

Week 4
Mon: 200π cm^3/s
Tue: $\frac{5}{6}$ ft/s
Wed: 2.025
Thu: Too high: $\sqrt[3]{x}$ is concave down there

Week 5
Mon: $\frac{1}{2}$
Tue: e
Wed: $c = \sqrt{3}$
Thu: $c = 2$

Week 6
Mon: $x = 0, \pm 2$
Tue: Local max ($f''(0) < 0$)
Wed: $x = \pm\frac{2}{\sqrt{3}}$
Thu: Max $\frac{1}{e}$ at $x = 1$; min 0 at $x = 0$

Week 7
Mon: 600 (a $10 \times 10 \times 10$ cube)
Tue: $t = 2$ and $t = 4$
Wed: 3
Thu: 24

Week 8
Mon: 20
Tue: 1
Wed: $3e^{9x^2}$
Thu: $\frac{1}{3}$

Week 9
Mon: $x^3 - 2\ln|x| + C$
Tue: $\frac{\sin^2 x}{2} + C$
Wed: $\frac{1}{2}\ln(x^2+1) + C$
Thu: $\ln\frac{3}{2}$

Week 10
Mon: 2
Tue: $\ln x + 1$
Wed: Concave down
Thu: $\frac{1}{2}$

Week 11
Mon: $e^x(x-1) + C$
Tue: $x \sin x + \cos x + C$
Wed: $x \ln x - x + C$
Thu: 1

Week 12
Mon: $\frac{1}{x} - \frac{1}{x+1}$
Tue: $\ln\left|\frac{x}{x+1}\right| + C$
Wed: $\ln|x-1| - \ln|x+4| + C$
Thu: $\ln\frac{4}{3}$

Week 13
Mon: 1
Tue: No, it diverges
Wed: 1
Thu: 2

Week 14
Mon: 0
Tue: 1.5
Wed: 2.5
Thu: Too low

Week 15
Mon: $y = \frac{1}{1-x}$
Tue: $y = 50e^{-0.2t}$
Wed: 500
Thu: 250

Week 16
Mon: $\frac{2}{\pi}$
Tue: 26
Wed: Displacement 0; distance 2
Thu: 26

Week 17
Mon: $\frac{8}{3}$
Tue: $\frac{9}{2}$
Wed: $\frac{\pi(e^2-1)}{2}$
Thu: $\frac{16}{15}$

Week 18
Mon: $\frac{\pi}{6}$
Tue: $\frac{13\pi}{15}$
Wed: $\frac{14}{3}$
Thu: $\int_0^1 \sqrt{1+4x^2}\,dx$

Week 19
Mon: 8
Tue: 13
Wed: $t = 10$ min
Thu: Total gallons in the first hour

Week 20
Mon: $-x\cos x + \sin x + C$
Tue: $\ln\left|\frac{x-1}{x+1}\right| + C$
Wed: $\frac{1}{2}$
Thu: 80

Week 21
Mon: $\frac{3t}{2}$
Tue: $y = 3x - 4$
Wed: $\frac{3}{4t}$
Thu: At $t = \pm 1$

Week 22
Mon: 5
Tue: π
Wed: $\frac{14}{3}$
Thu: Distance 2π; displacement 0

Week 23
Mon: $\langle 2t, e^t \rangle$
Tue: 3
Wed: $\langle 6, 0 \rangle$
Thu: $\langle 5, 5 \rangle$

Week 24
Mon: $x^2 + y^2 = 2y$
Tue: Toward ($\frac{dr}{d\theta} = -2$)
Wed: $-\frac{2}{\pi}$
Thu: $\frac{\pi}{4}$, $\frac{3\pi}{4}$

Week 25
Mon: 9π
Tue: $\frac{3\pi}{2}$
Wed: $\frac{\pi}{8}$
Thu: 3π

Week 26
Mon: 3
Tue: 3
Wed: $\frac{5}{3}$
Thu: 1

Week 27
Mon: No: the terms approach 1
Tue: Yes ($p = 1.5 > 1$)
Wed: No
Thu: No

Week 28
Mon: Yes (compare with $\frac{1}{n^2}$)
Tue: No (compare with $\frac{1}{n}$)
Wed: No
Thu: Yes

Week 29
Mon: Yes, conditionally
Tue: Yes
Wed: $\frac{1}{5}$
Thu: $\frac{2}{3}$

Week 30
Mon: $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$
Tue: $(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$
Wed: 0.995
Thu: At most 0.0002

Week 31
Mon: 3
Tue: $[-1, 1)$
Wed: $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$
Thu: $x + x^3 + \frac{x^5}{2}$

Week 32
Mon: $\frac{1}{3}$
Tue: $x^x(\ln x + 1)$
Wed: $e^x(x^2 - 2x + 2) + C$
Thu: 1

Week 33
Mon: Conditionally
Tue: 3
Wed: π

Week 34
Mon: 2
Tue: 1

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